

SIMATS ENGINEERING

COURSE NAME: Advance Wireless Communication Systems /
5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 3 - Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems. (BL6)

Assessment Tool Number	: CO3_AT1
Assessment Tool Weightage	: 35%
Assessment Tool	: Design Task
Duration	: 180 Minutes
Marks	: 100 Marks
Type of Assessment Conduction	: At Home
Evaluation Type	: GitHub
Result Generation	: Students design and justify appropriate wireless communication solutions by applying engineering principles to meet specified requirements.
Topic Covered	: Spectrum management, licensed/unlicensed spectrum, dynamic spectrum access, cognitive radio, OFDMA, resource block allocation, GFDMA, NOMA, power-domain multiplexing, SIC, and IoT applications. Advanced waveforms and wireless technologies including GFDM, FBMC, UFMC, mmWave propagation, path loss and penetration, millimeter-wave antenna systems, MIMO, and Massive MIMO fundamentals.

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills.

QUESTIONS: 1 (1 X 100 = 100)

Total Marks: 100 Duration: 3hrs

11. Design a virtualized wireless access framework for next-generation communication.

Code of Conduct:

I G. Lakshmi Kanth (192312411) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

G. Lakshmi Kanth

Signature of the student: *G. Wibe*

RUBRICS FOR CALCULATION:

Design Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides indepth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback
Suggestions for Improvement	20%	Offers innovative, practical, and well-justified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided
Communication	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communication and minimal participation

Design of a virtualized wireless Access framework for next Generation communication.

A virtual wireless access framework separates network hardware from software, allowing network functions to run on virtual machines or containers instead of dedicated equipment.

Working:

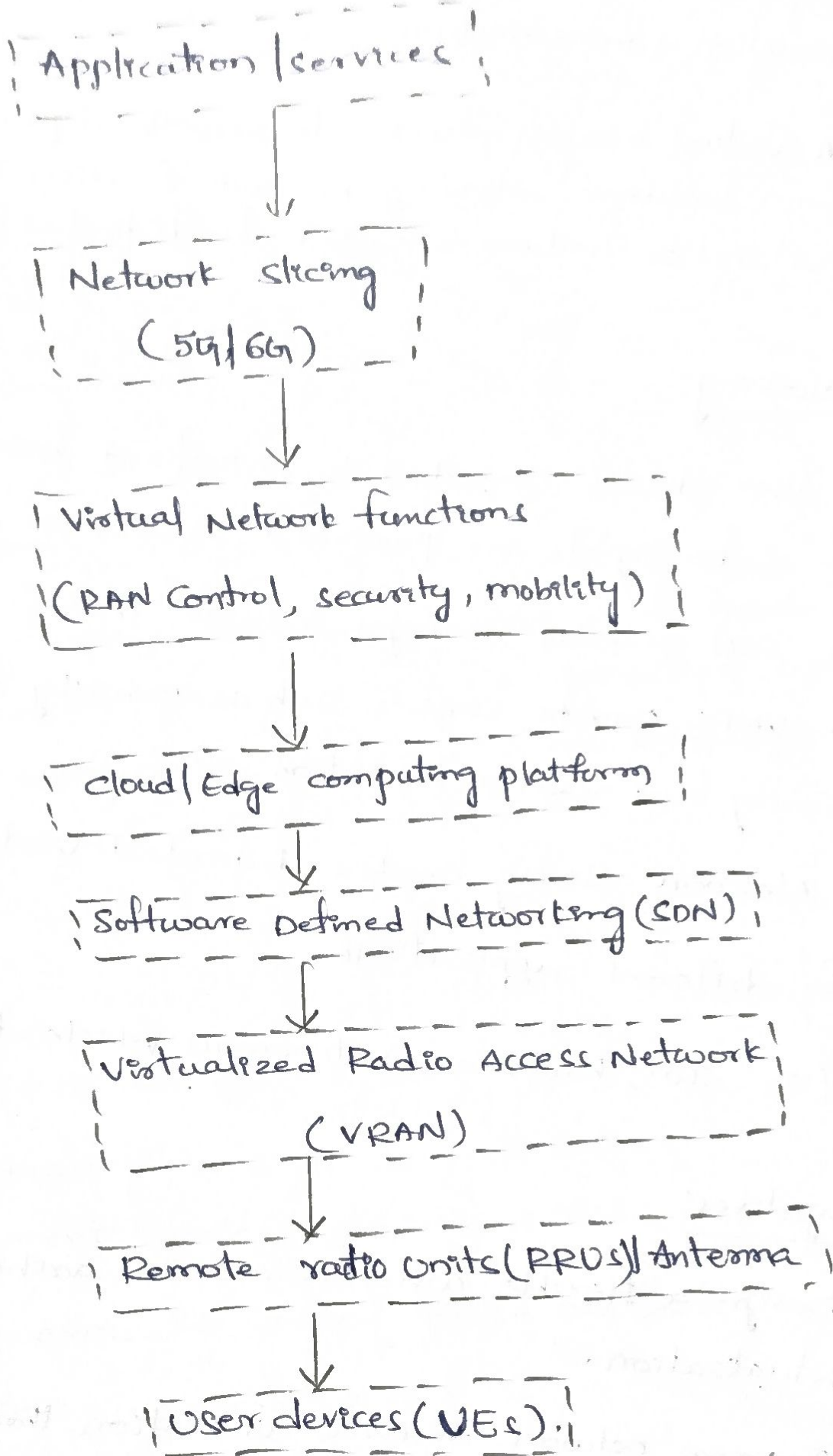
- 1) User devices connected to virtualized base stations.
- 2) Radio signals are processed in cloud or edge servers.
- 3) SDN controller manages network traffic dynamically.
- 4) VNFS provide services such as mobility management, security and resource allocation.
- 5) Network slicing creates dedicated virtual networks for different applications.

(Eg: IOT, healthcare, autonomous vehicle etc..).

Objectives:

- 1) Design a flexible wireless access architecture using virtualization.
- 2) Improve network resource utilization through cloud based processing.
- 3) Reduce hardware deployment and maintenance costs.
- 4) minimize latency using edge computing.

Flow diagram



System architecture:

- 1) User Equipment (UE)
- 2) Virtualized Radio access network (VRAN)
- 3) Software Defined Networking (SDN)
- 4) Network function virtualization (NFV)
- 5) Edge and cloud computing
- 6) Network slicing.

Matlab based simulation:

The simulation displays CPU usage using a bar graph, throughput using a line graph, and resource utilization using a pie chart. The average throughput value is also calculated to evaluate network performance.

Advantages:

- * Efficient utilization of computing resources.
- * Reduced operational and maintenance costs.
- * Easy deployment of new network services.
- * Supports dynamically scaling based on traffic demand.
- * Low latency through edge computing.
- * High reliability and improved quality of service.
- * Better energy efficiency.
- * Simplified network management.

Applications:

- 1) 5G, 6G future communication system.
- 2) Industrial automation and Industry 4.0.
- 3) IoT
- 4) Smart agriculture.
- 5) cloud gaming and multimedia streaming.
- 6) Intelligent transportation systems.

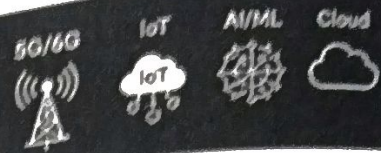
Conclusion:

Virtualized wireless Access frameworks represent the future of wireless communication by replacing hardware based network functions with software defined virtual services.

The integration of SDN, NFV, edge computing, cloud computing and network slicing provides a flexible, scalable and cost effective communication system.

DESIGN OF A VIRTUALIZED WIRELESS ACCESS FRAMEWORK FOR NEXT-GENERATION COMMUNICATION

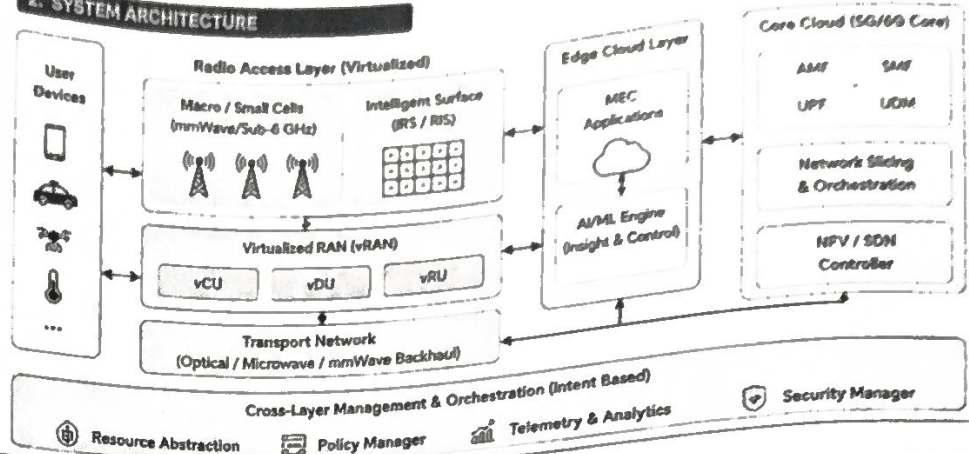
Goal: Provide intelligent, flexible and efficient wireless access using virtualization, cognitive spectrum access and advanced waveforms for 5G/6G and IoT services.



1. DESIGN OBJECTIVES

- Efficient spectrum utilization through cognitive and dynamic access.
- Support multiple services (eMBB, URLLC, mMTC).
- Virtualize RAN and core functions for elasticity and slicing.
- Support advanced waveforms and multiple access techniques.
- Enable edge intelligence for low latency.
- Scalable, secure and energy-efficient architecture.

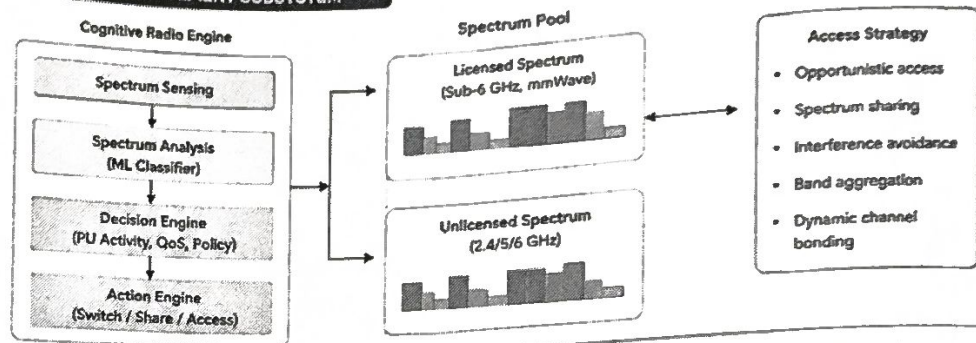
2. SYSTEM ARCHITECTURE



3. KEY FEATURES

- Cognitive Spectrum Access**
Sense - Learn - Decide - Act loop for dynamic access to licensed/unlicensed spectrum.
- Virtualization & Slicing**
NFV/SDN based vRAN and core with end-to-end network slicing for diverse use cases.
- Advanced Waveforms & Multiple Access**
Support OFDMA, NOMA, GFDM/FBMC for high spectral efficiency.
- Edge Intelligence**
AI/ML for traffic prediction, user clustering, mobility and interference management.
- Scalability & Flexibility**
Elastic resource scaling on demand in cloud and edge.
- Security by Design**
End-to-end authentication, isolation between slices, and secure spectrum access.

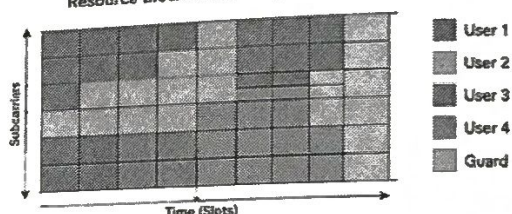
4. SPECTRUM MANAGEMENT SUBSYSTEM



5. MULTIPLE ACCESS & WAVEFORM LAYER

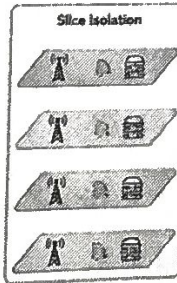
Technique	Purpose / Benefit
OFDMA	Orthogonal resource allocation for multi-user fairness and high throughput.
NOMA (Power Domain)	Support massive connectivity with higher spectral efficiency using SIC.
GFDM / FBMC	Better spectral containment, lower out-of-band emission, suitable for fragmented spectrum.
UFMC	Low complexity alternative with good performance.

Resource Block Allocation (Example - OFDMA)



6. NETWORK SLICING

Use Case	Slice Characteristics
eMBB Slice	High throughput, wide bandwidth, GFDM/OFDMA, high priority.
URLLC Slice	Ultra-low latency (<1 ms), NOMA/OFDMA, edge processing.
mMTC Slice	Massive connections, NOMA, power saving, low data rate.
IoT Slice	Long battery life, narrowband IoT, low complexity.



7. SIGNAL/PROPAGATION CONSIDERATIONS (mmWave)

Path Loss Model (Example)

$$PL(dB) = 32.4 + 20 \log_{10}(f_{GHz}) + 20 \log_{10}(d_{km}) + X_d$$

Where: f = frequency (GHz), d = distance (km), X_d = shadowing (dB)

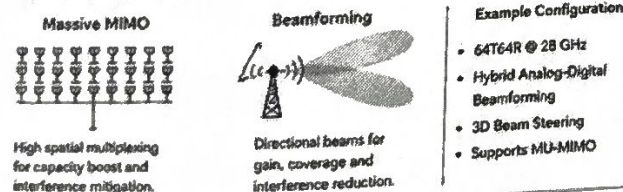
Frequency (GHz)	Path Loss @ 100 m (dB)
28	-101 dB
60	-115 dB
100	-120 dB

Mitigation: Beamforming, Massive MIMO, IRS, Small Cells, Multi-connectivity.

Penetration Loss (Typical)

Material	Loss @ 28 GHz
Concrete Wall (20 cm)	20 - 30 dB
Glass	2 - 5 dB
Wood Door	3 - 6 dB
Brick Wall	10 - 15 dB

8. MIMO & ANTENNA SYSTEMS



9. SECURITY & RELIABILITY



11. EXPECTED OUTCOMES

- Higher spectral efficiency and better user experience.
- Support for diverse 5G/6G and IoT services through slicing.
- Reduced latency using edge intelligence.
- Efficient spectrum use via cognitive access.
- Scalable, flexible and cost-effective network operation.

12. CONCLUSION

The proposed virtualized wireless access framework integrates cognitive spectrum management, advanced waveforms, virtualization, edge intelligence and Massive MIMO to deliver a flexible, efficient and future-ready solution for next-generation communication.

10. IMPLEMENTATION VIEW (TOOLS & TECHNOLOGIES)



Simulation: MATLAB/Simulink, ns-3, QuaDRiGa
Deployment: Cloud (OpenStack/K8s), Edge (KubeEdge)

```

clc;

clear;

close all;

% Number of users
users = 10;

% Total available bandwidth (MHz)
total_bw = 100;

% Generate random user demand (Mbps)
demand = randi([5 20],1,users);

% Allocate equal bandwidth
allocated_bw = total_bw/users;

% Calculate throughput
throughput = allocated_bw .* (demand/max(demand));

% Display results
disp('User Demand(Mbps) Allocated BW(MHz) Throughput(Mbps)');

for i = 1:users

    fprintf('%2d\t\t%2d\t\t%5.2f\t\t%5.2f\n', ...
        i,demand(i),allocated_bw,throughput(i));

end

% Plot Throughput

figure;

bar(throughput);

xlabel('Users');

ylabel('Throughput (Mbps)');

title('Virtualized Wireless Access Framework');

grid on;

```

